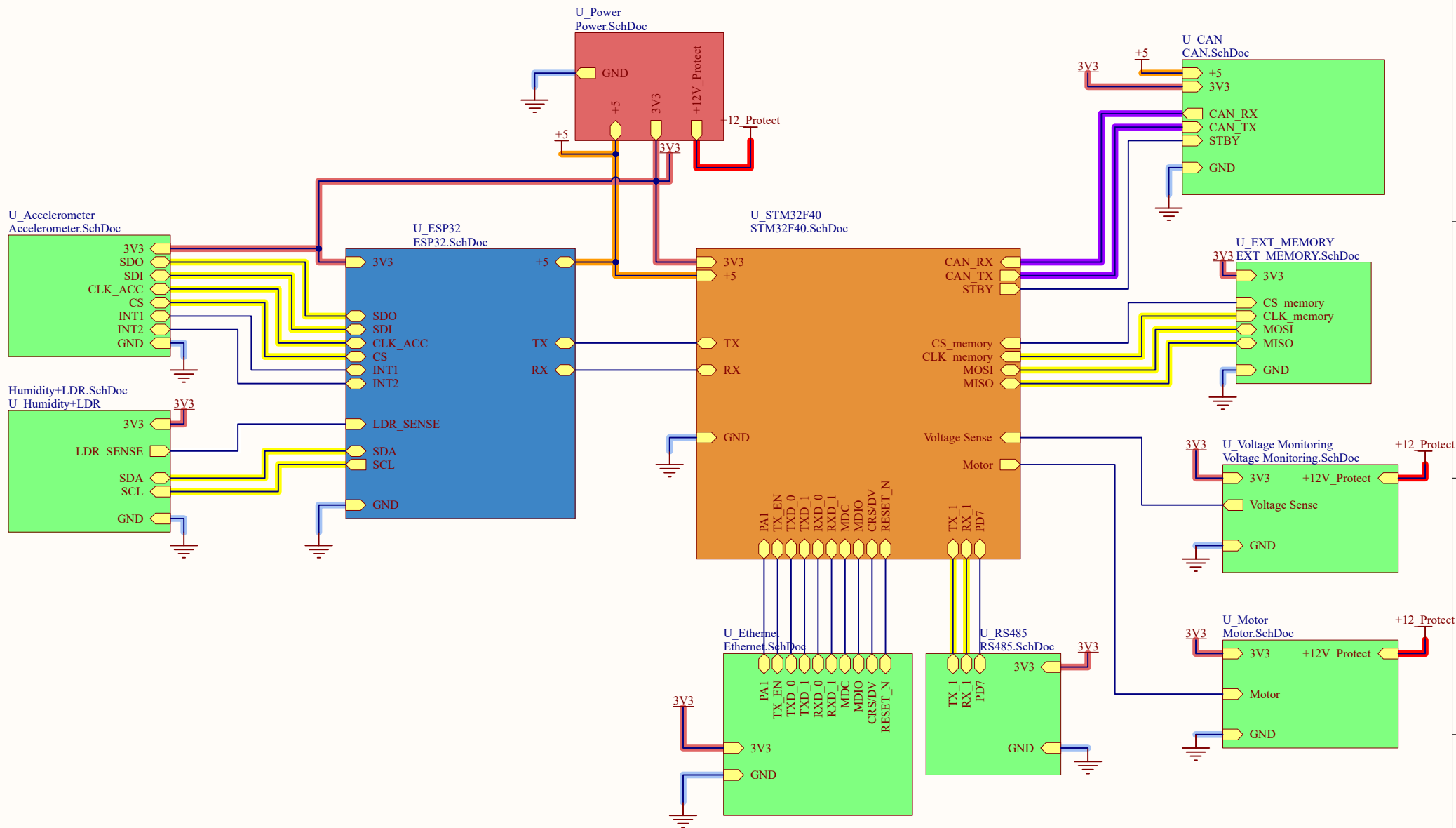
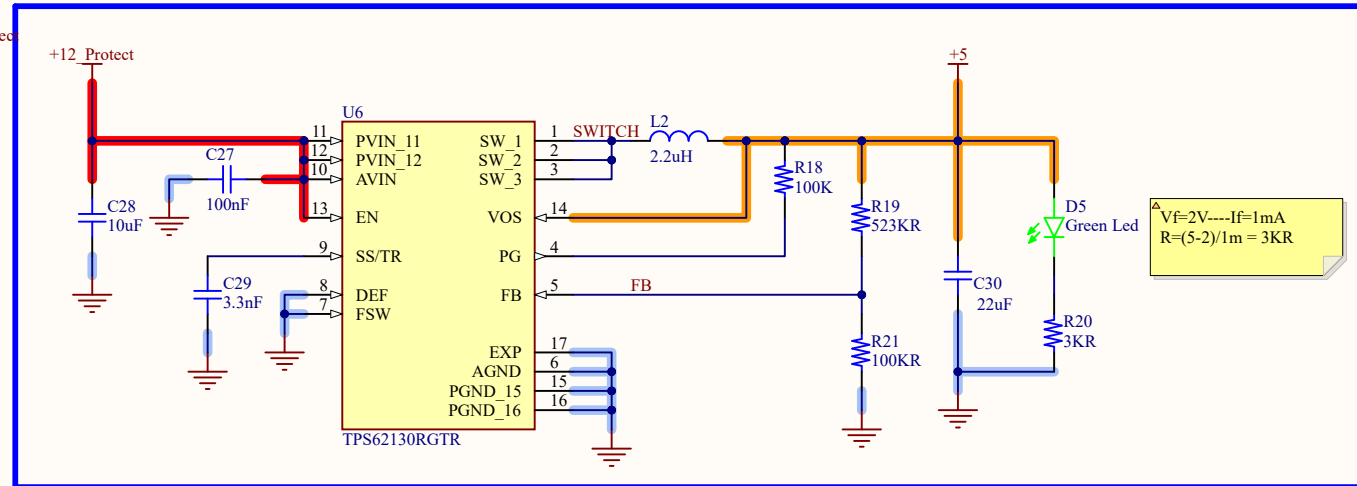
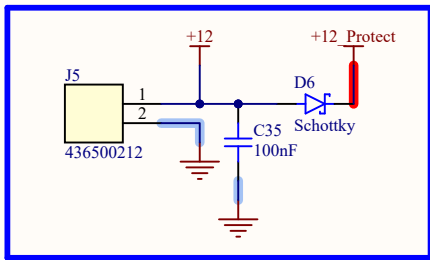




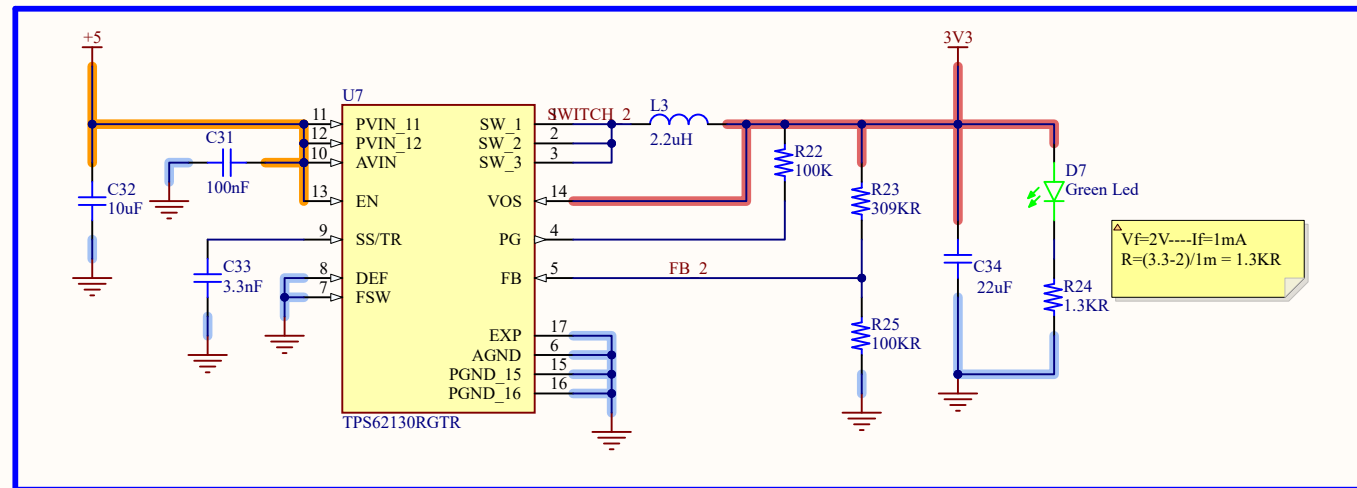
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ESP32---STM32 BOARD		
Size	Number	Revision
A4		
Date:	7/06/2026	Sheet of
File:	D:\Hardware projects\...\Master.SchDoc	Drawn By: MOHAMAD ASHOUR

12V >>>>> 5V

I/P Connector

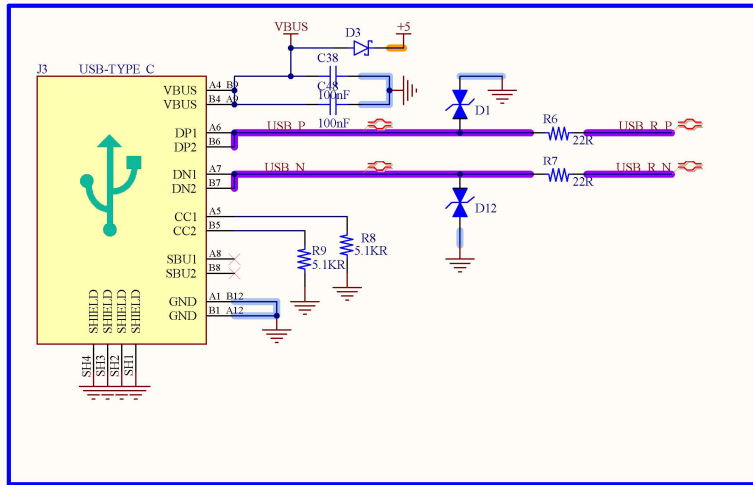


5V >>>>> 3V3

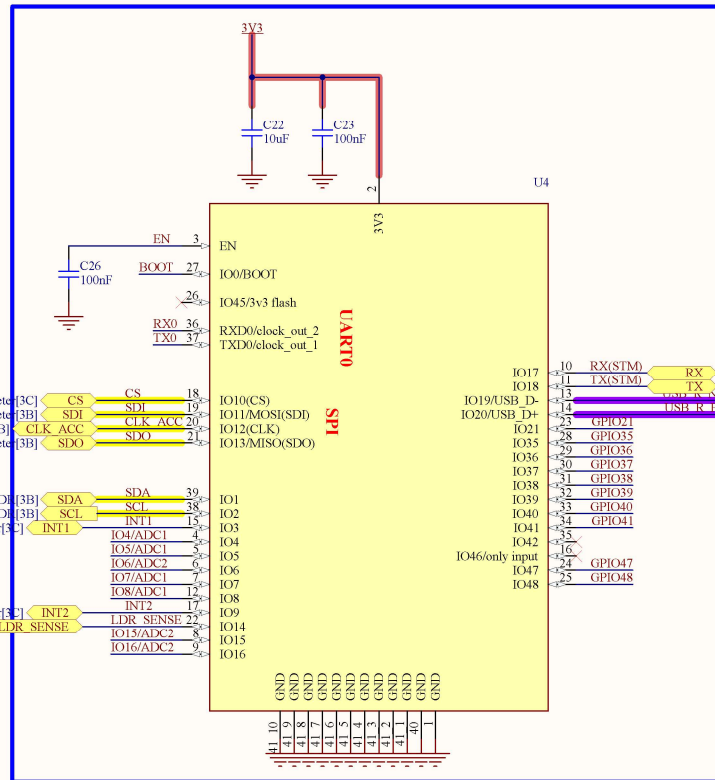


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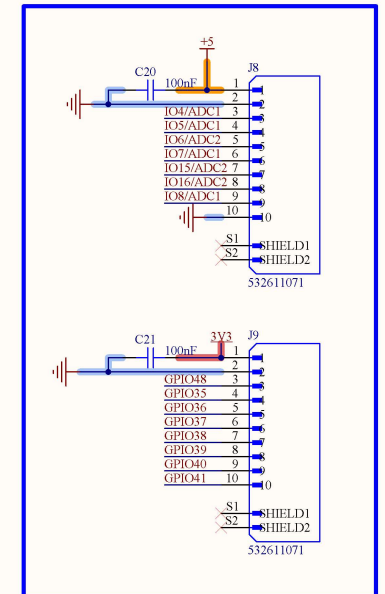
USB_C



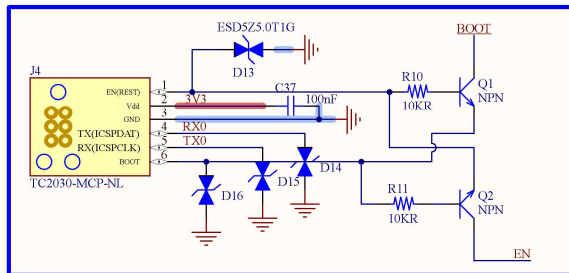
ESP32_S3_WROOM1



Additional Pins

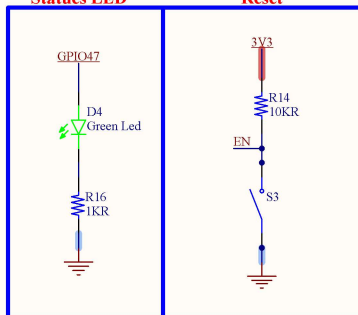


Tag Connector

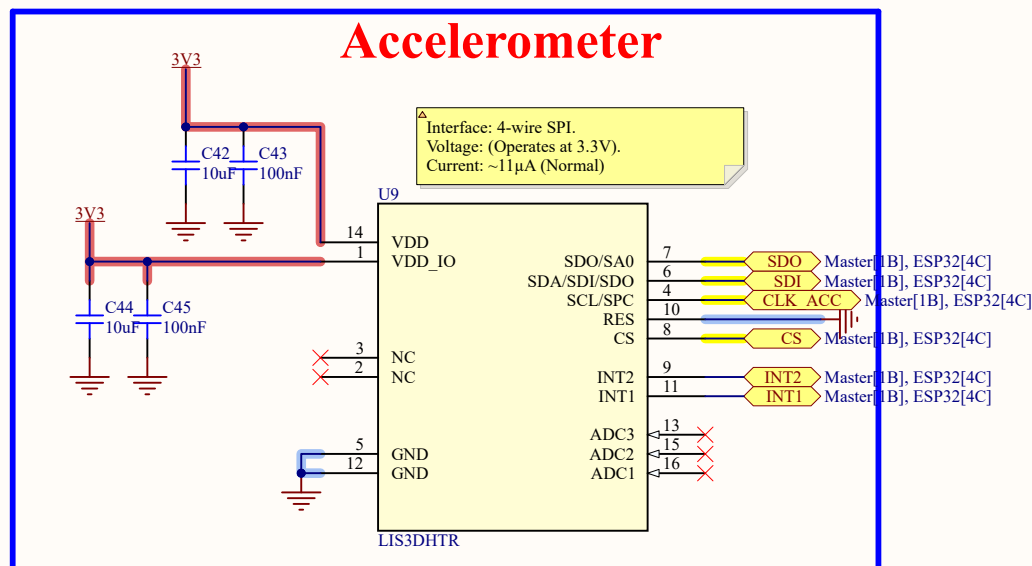
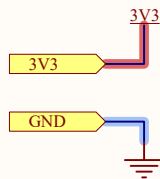


Status LED

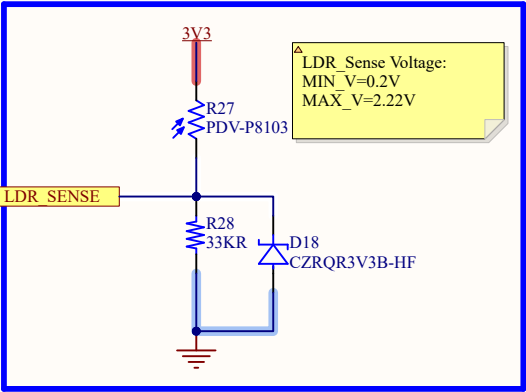
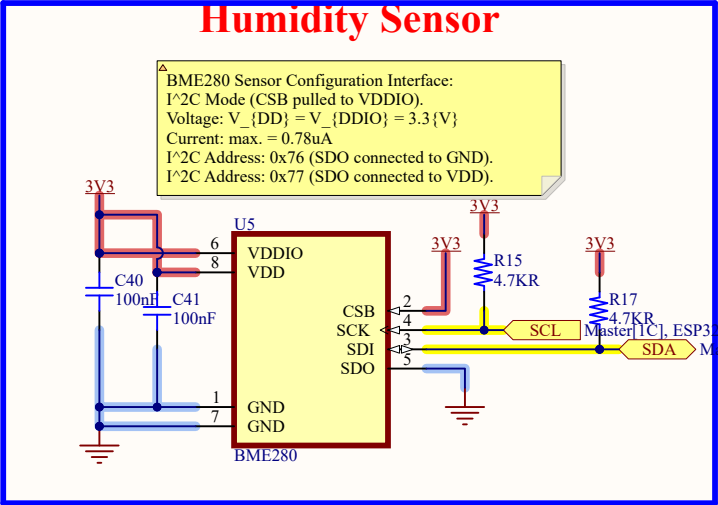
Reset



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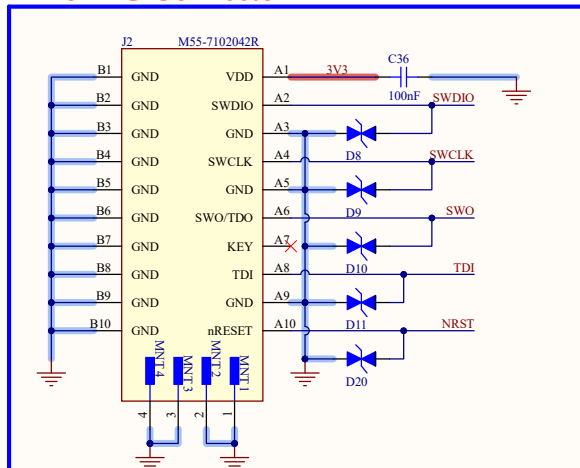


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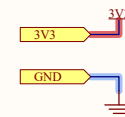
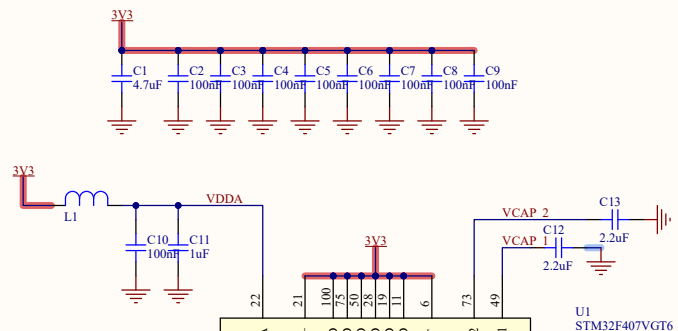


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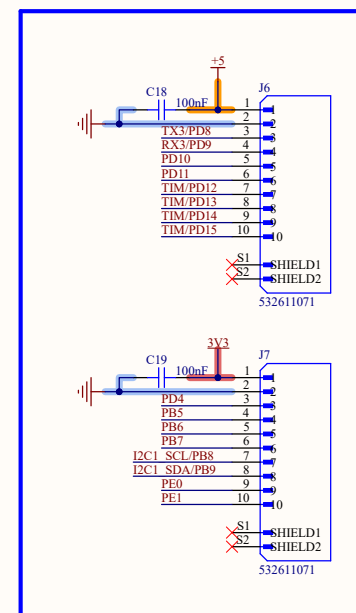
JTAG Connector



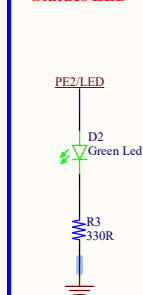
STM32



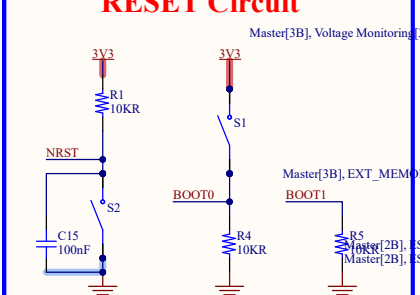
Additional Pins



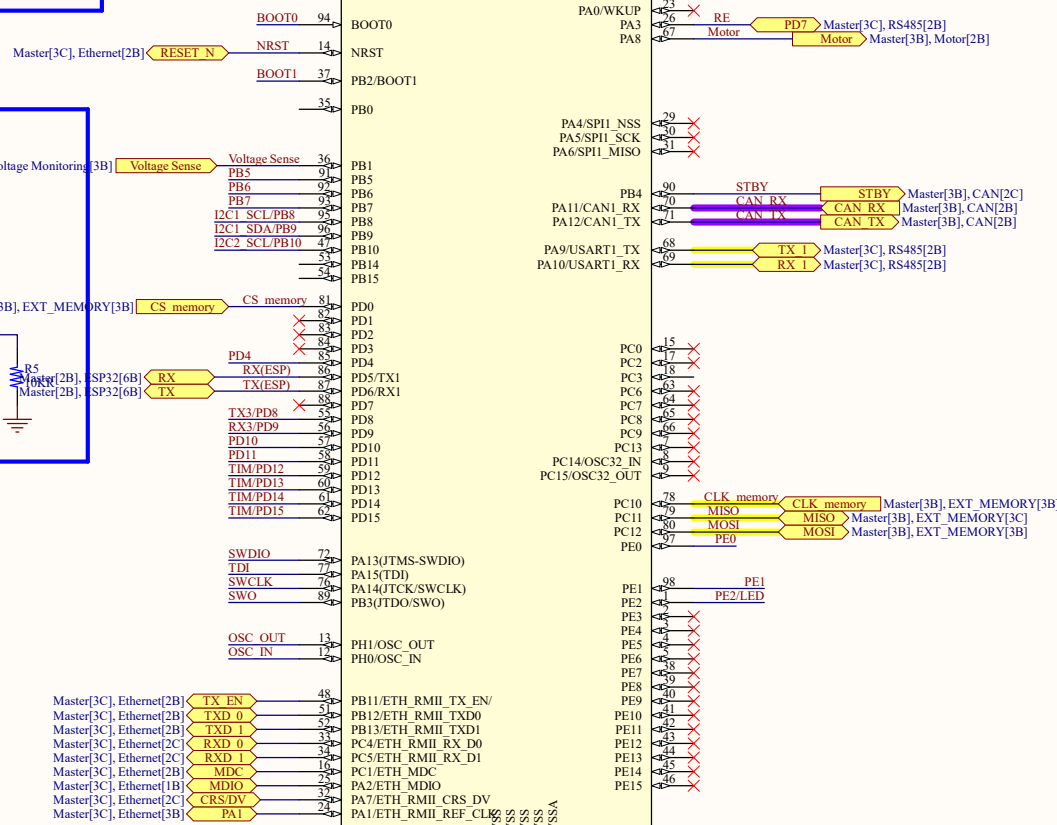
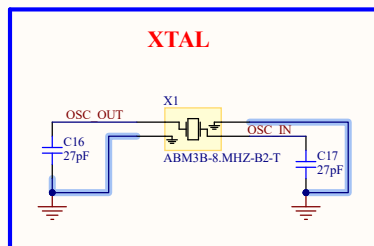
Status LED



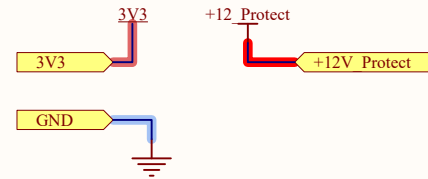
RESET Circuit



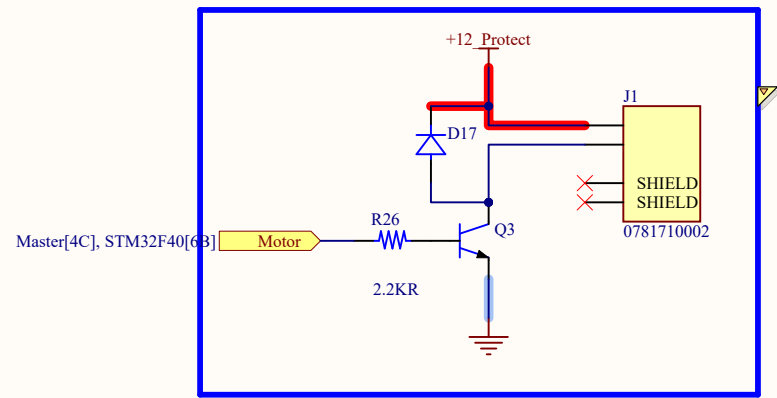
XTAL



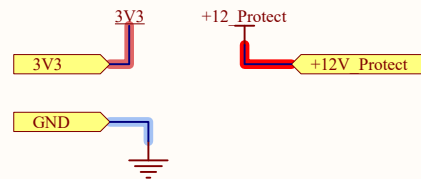
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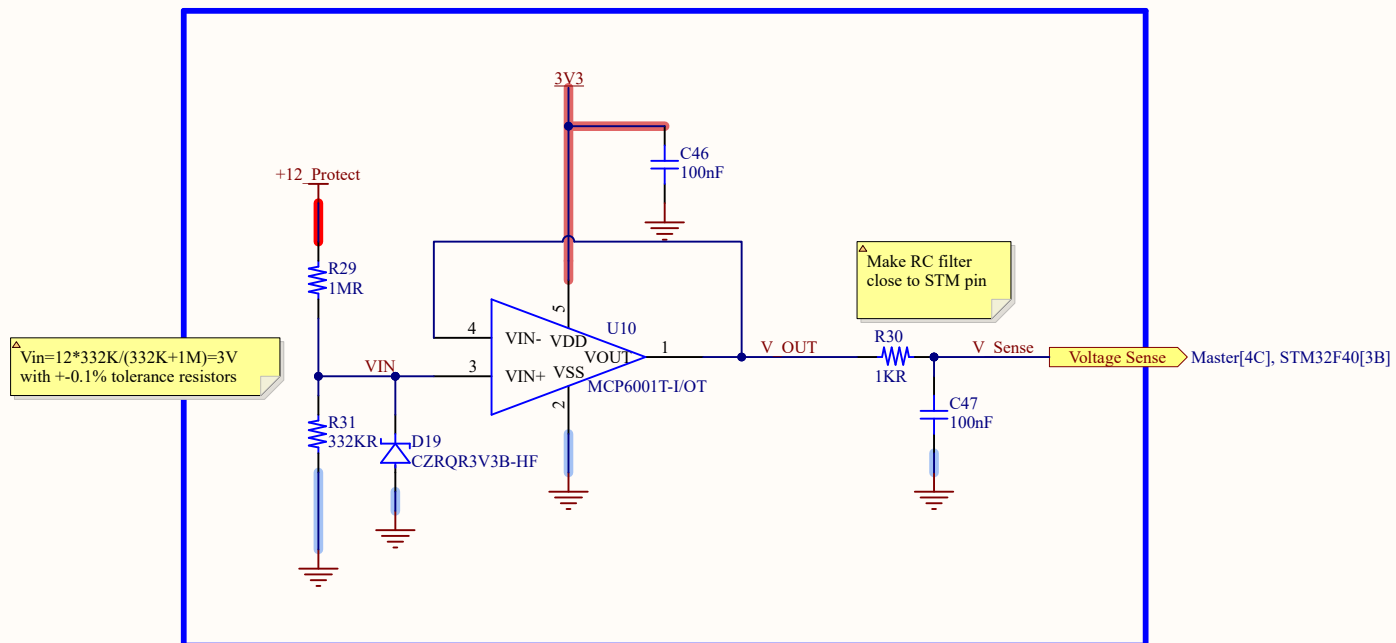
Motor Controller Circuit



Title		
Size A4	Number	Revision
Date:	7/06/2026	Sheet of
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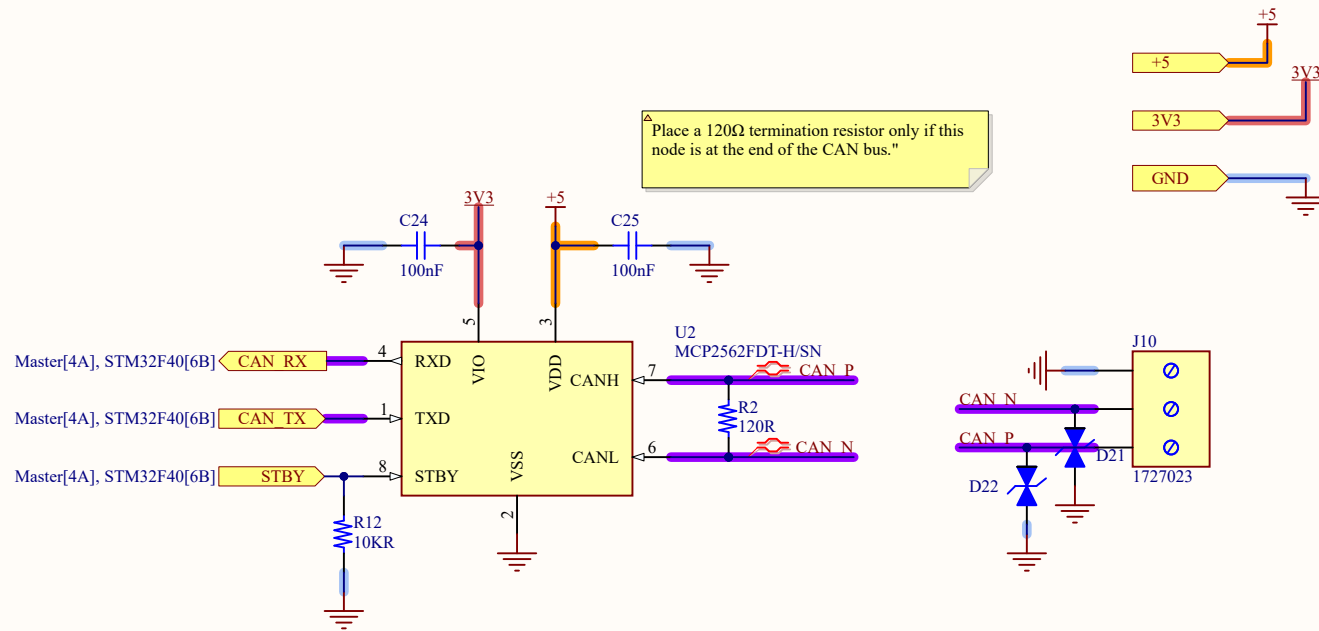


Voltage Monitoring



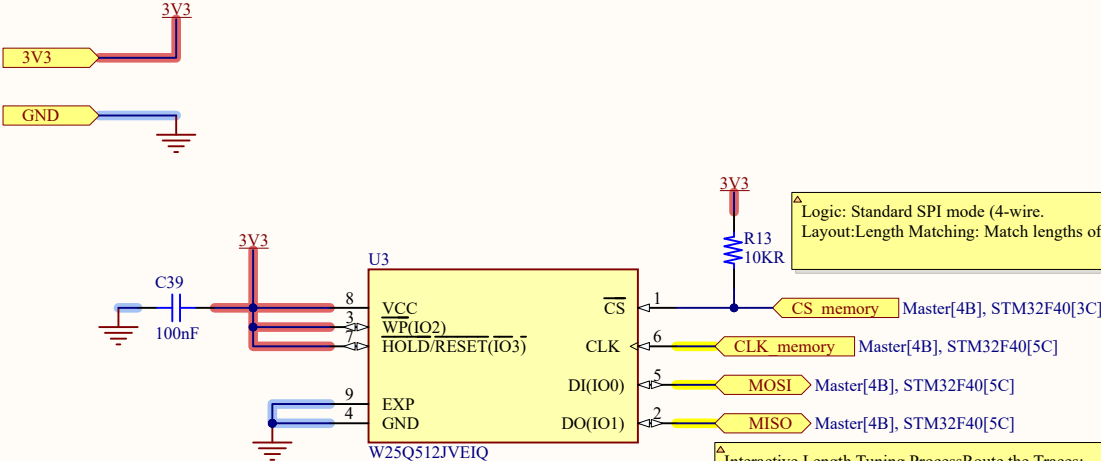
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Date: 7/06/2026	Sheet of	
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CAN



Title		
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A4		
Date:	7/06/2026	Sheet of
File:	D:\Hardware projects\...\CAN.SchDoc	Drawn By:

EXT.Memory

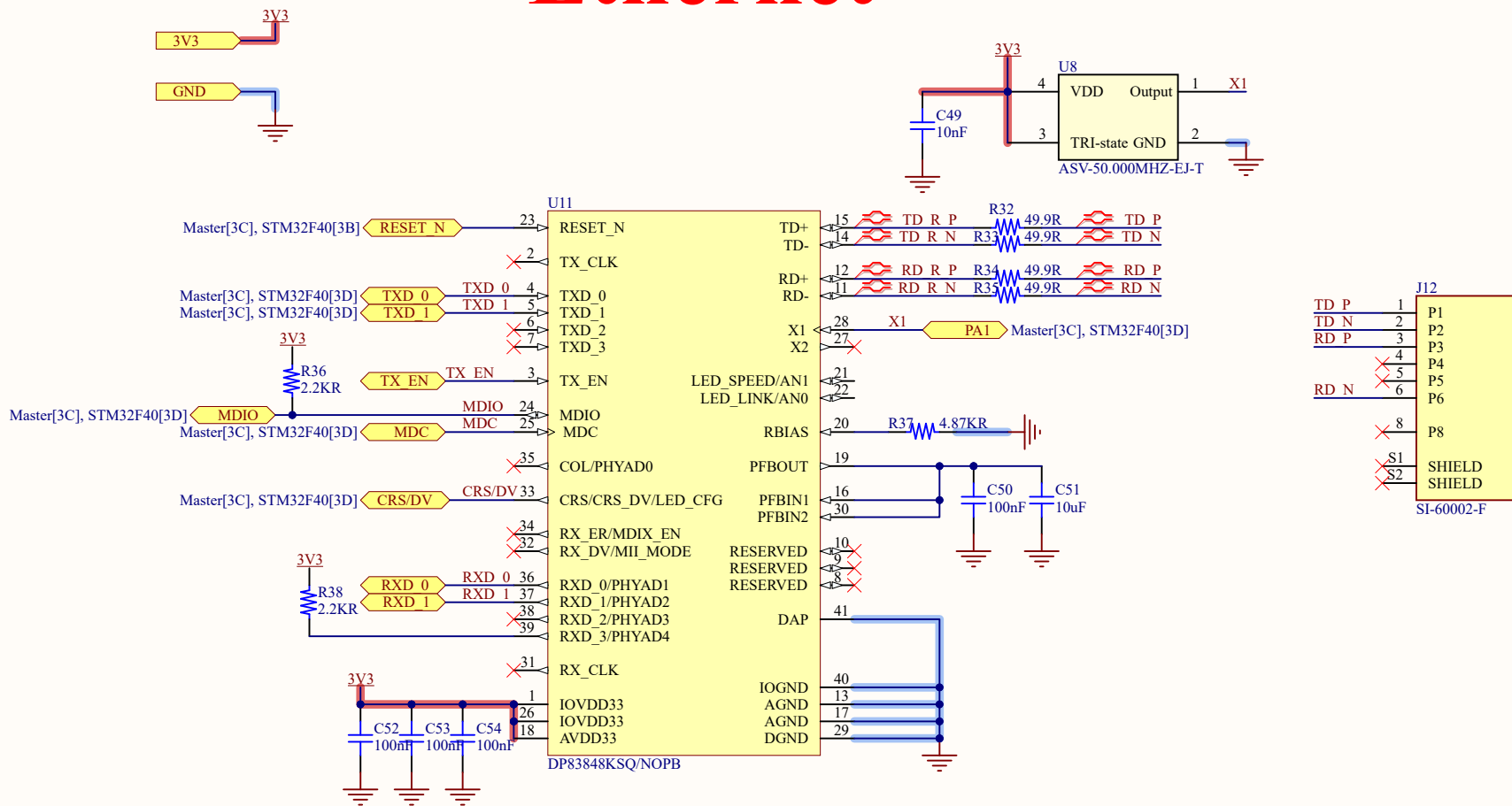


Logic: Standard SPI mode (4-wire).
Layout: Length Matching: Match lengths of SCK, MISO, and MOSI traces(+/- 1 mm).

Interactive Length Tuning Process
Route the Traces:
Begin by routing SPI signals (SCK, MISO, MOSI) normally from the MCU to the Flash memory.
Identify the Reference: Determine which of the three traces is the longest; this becomes your target length (reference).
Apply Meandering: Use the Length Tuning tool to add "Squiggles" to the shorter traces until their total length matches the reference within a tolerance of (+/- 1 mm).

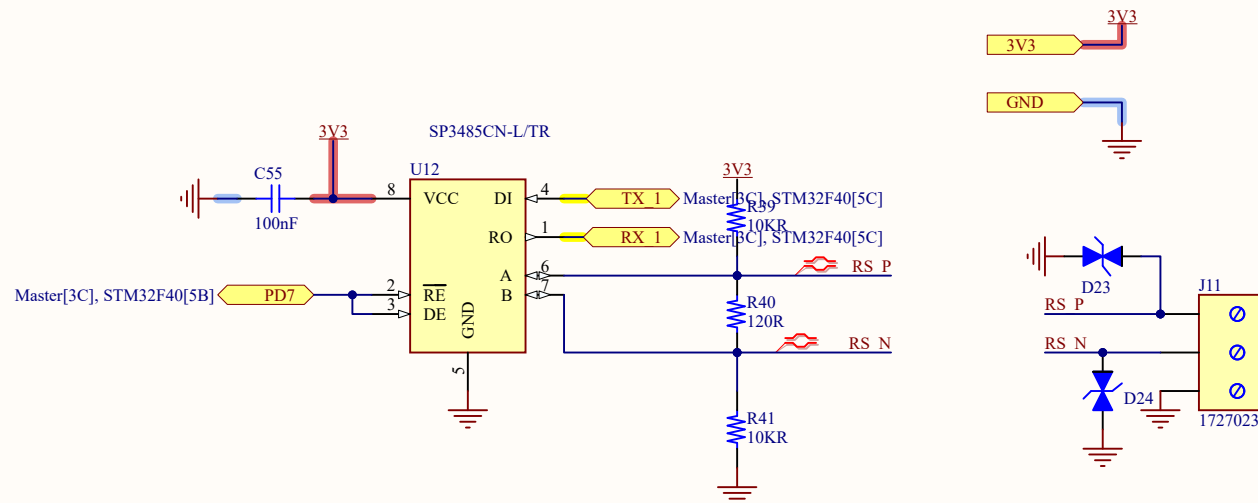
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Size	Number	Revision
A4		
Date:	7/06/2026	Sheet of
File:	D:\Hardware projects\..EXT MEMORY\Sheet By:	

Ethernet



Title		
Size A4	Number	Revision
Date:	7/06/2026	Sheet of
File:	D:\Hardware projects\Ethernet.SchDoc	Drawn By:

RS485



Title		
Size A4	Number	Revision
Date:	7/06/2026	Sheet of
File:	D:\Hardware projects\..RS485.SchDoc	Drawn By: